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Integrated Sensing and Communication in Wi-Fi and Cellular: An Industry and Standards Perspective

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(Invited Article)

ABSTRACT Integrated sensing and communication (ISAC) is expected to be a key enabling technology for diversifying the use cases of future telecommunication systems. Recently, there has been an active push in the industry on ISAC, with several strides being made in standardization efforts in both Wi-Fi and cellular communication domains. This article provides a comprehensive overview of the current efforts toward ISAC in Wi-Fi and cellular communication (5G and 6G), highlighting current research directions and standardization efforts. The article also discusses the current advances in channel modeling used for evaluating ISAC systems and the impact of the different transceiver hardware impairments on ISAC. We also discuss the limitations of current progress and summarize key research directions for the future.

INDEX TERMS 3rd generation partnership project (3GPP) standards, 5G, 6G, IEEE Standards, integrated sensing and communication (ISAC), Wi-Fi, wireless sensing.

INTRODUCTION

Wireless sensing, which involves the use of radio waves to detect and monitor environmental parameters in the surroundings of the transmitter (TX) or receiver (RX) of the waves, has been around for almost a century. It started with the development of radar and sonar technologies [1], primarily for military applications, and then expanded to other applications, including navigation, remote sensing, telemetry, weather forecasting, localization, tracking, etc. However, the need for dedicated and expensive hardware for sensing has kept the technology away from the mass consumer market space. On the other hand, with the exponential growth of consumer electronics, there is a ubiquitous availability of wireless communication devices today. In fact, the demand for higher spectral efficiency has led to sophisticated communication hardware being deployed that can support higher bandwidths, millimeter-wave (mm-wave) frequencies, and multiple input multiple output (MIMO) technology. While supporting higher spectral efficiency, such hardware can also provide good range and angular resolution, which are key requirements for wireless sensing [2]. This has organically led to the goal of repurposing communication hardware to also perform sensing, a concept called integrated sensing and communication (ISAC) [3]. ISAC involves the use of a communication TX to transmit a sensing signal, the use of the received signal at a RX to estimate variations in the channel between TX and RX, and the performance of sensing operations based on the variations, as depicted in Fig. 1. By reducing barriers for the use of sensing, ISAC promises to make sensing available to the mass consumer market.

The concept of ISAC represents a significant evolution in wireless technology, merging the capabilities of traditional telecommunications networks with advanced sensing technologies such as radar. The benefits are twofold:

1) With the saturation of the communication requirements of many applications and the subscriber count, the newly added sensing functionality can spur the demand for next-generation communication hardware and

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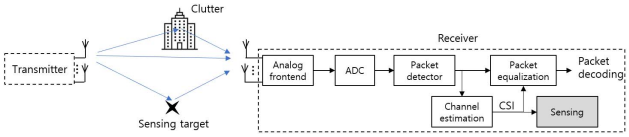


FIG. 1. Illustration of the signal transmission, reception and processing to enable sensing with communication hardware.

provide new monetization opportunities. This is particularly true in the case of mm-wave chips, which were found to have limited benefits for communication due to the sporadic connectivity.

2) The use of the widely deployed communication hardware for sensing enables ubiquitous sensing coverage.
3) The use of common hardware, and potentially common signals, for communication and sensing improves spectral efficiency and reduces hardware cost.

It should be noted that some types of sensing are inherently performed by conventional telecommunication devices to improve communication, such as channel estimation and beamforming training. ISAC also promises to further improve these features [4]. On the flip side, repurposing communication hardware for sensing poses challenges related to hardware impairments, signal interference, and the need for new standards and protocols to support ISAC, as shall be described later.

The use of ISAC has been explored for several different communication protocols, listed as follows.

1) *Wi-Fi*: It access points (APs) and client devices can be equipped with sensing capabilities to realize applications for smart home, indoor navigation, health monitoring, etc. [5], [6]. Due to their low hardware cost and short communication range, Wi-Fi devices are typically used for cost-sensitive short-range sensing applications.
2) *Cellular*: Sensing functions can be integrated in cellular systems, which can be equipped on both a base station (BS) and a user equipment (UE), to facilitate multiple sensing modes. This integration can improve network performance by providing real-time environmental data and also facilitate new applications such as smart factory, weather monitoring, smart transportation, public safety [2], [3], [7], [8], etc. The evolution of 5G Advanced (5G-A) networks and the development of 6G technologies are expected to further spur this integration of communication and sensing. Due to their larger array sizes, better synchronization, and wider communication range, cellular sensing is typically used for long-range or large-coverage sensing applications.

It should be noted that Bluetooth and ultrawide band (UWB) are other potential candidate ISAC technologies. However, there is very limited work on sensing with Bluetooth [9], due to its simpler hardware and frequency-hopping protocol. Although UWB is an ideal candidate for sensing because of its large bandwidth, it has seldom been used for communication. Consequently, for the sake of brevity, we skip these two technologies from this discourse.

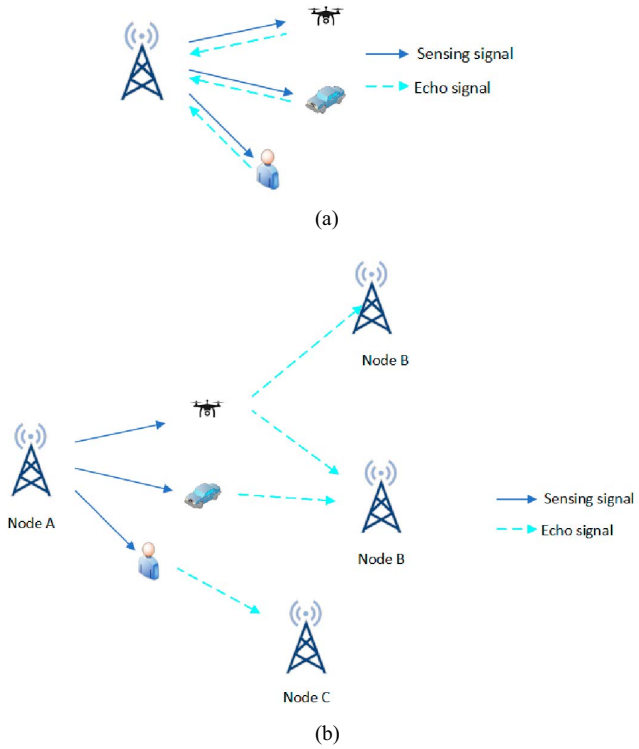


FIG. 2. Illustration of different sensing modes in ISAC where the TX and RXs are BSs. In general, the TX and RXs may also be Wi-Fi access points or UEs. (a) Monostatic sensing. (b) Multistatic sensing.

Different types of sensing mechanisms have been considered in ISAC including: 1) monostatic sensing, where the sensing TX and the sensing RX are on the same device; 2) bistatic sensing, where the sensing TX and the sensing RX are noncollocated; and 3) multistatic sensing, which is a generalization of bistatic sensing involving multiple sensing RXs, as depicted in Fig. 2. Monostatic sensing is not plagued by hardware impairments such has oscillator mismatch or timing errors but requires strong self-interference cancelation mechanisms to cancel the interference from TX to RX. Bistatic and multistatic sensing do not suffer from strong self-interference but demand stricter hardware requirements and mechanisms for TX-RX synchronization over the air or via another medium. The use of ISAC has been considered both at sub-7 GHz frequencies [referred to as wireless local area network (WLAN) in Wi-Fi and FR1 in cellular] and at mm-wave frequencies [referred to as directional multi-gigabit (DMG) in Wi-Fi and FR2 in cellular]. While the former offers wider coverage, the latter offers better angular and range resolution.

Taking advantage of the ubiquitous availability of communication hardware, many different use cases have been explored for ISAC. Based on the coverage area, these can be broadly classified into three groups.

1) *Personal Area Sensing*: These include use cases where the hardware is carried by an individual and/or is present in close proximity to the individual. This includes use cases such as gesture recognition, activity detection, and user pose estimation, and medical diagnostics [5], [6], e.g., breathing rate monitoring, heart rate monitoring, and sleep stage detection.

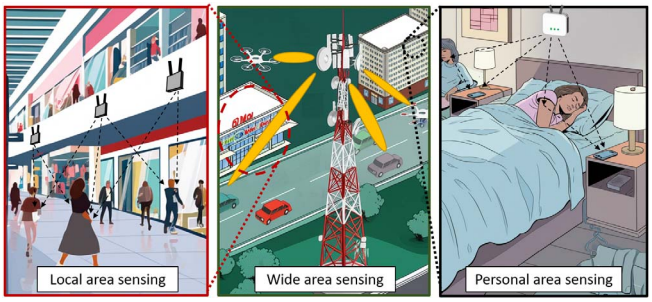


FIG. 3. Illustration of the three classes of ISAC use cases based on coverage range.[1]

2) *Local Area Sensing*: These include use cases where the sensing area includes a room, residential area, an office building, etc. It includes use cases such as localization and tracking, proximity detection, fall detection, simultaneous localization and mapping, people counting, and intruder detection [6], [10].
3) *Wide Area Sensing*: These include use cases where the sensing area includes a larger coverage area such as a campus, a mega-construction, a base-station coverage region, and a road network. It includes use cases such as vehicle or unmanned aerial vehicle (UAV) tracking/localization, detecting cracks in buildings, large-scale logistics, animal migration monitoring, and weather monitoring [8].

Some examples of these three classes of use cases are illustrated in Fig. 3. In fact, 32 use cases have been listed in 3rd generation partnership project (3GPP) Release 19 [8], including smart home, smart factory, smart transportation, etc. On the Wi-Fi side, in the IEEE 802.11bf standard, key use cases broadly cover room sensing, gesture recognition, health care, 3-D vision, and in-car sensing applications [11].

On the Wi-Fi side, some papers [12] have summarized well the contributions of the IEEE 802.11bf standard to enable sensing [13]. However, the impact of ongoing standards such as IEEE 802.11bn and IEEE 802.11bq on ISAC has not been discussed. There is also limited work summarizing the channel models used to evaluate sensing with Wi-Fi [14], [15] and the impact of hardware impairments on Wi-Fi sensing [16]. On the cellular side, several papers have well summarized the identified ISAC use cases and solutions with applications to 5G-A and future 6G [3], [7], [8]. However, there is limited work summarizing current efforts in 3GPP toward 5G-A and the future 6G standard, and the channel models being considered. This article provides an overview of industry standardization efforts, their limitations, and the expected progress in the near future in Wi-Fi and cellular ISAC technologies. In addition, we also discuss the channel models proposed for validating ISAC use cases and the impact of communication hardware impairments on sensing.

The organization of the article is as follows. In "ISAC in Wi-Fi" section of this article, we focus on ISAC in Wi-Fi; in "Progress in Wi-Fi Sensing" section, we discuss existing work on Wi-Fi sensing, relevant standards and future directions; in "Channel Models for Wi-Fi Sensing" section, we discuss channel modeling for Wi-Fi sensing; and in "Modeling of Hardware Impairments" section, we discuss the modeling of hardware impairments which can impact Wi-Fi sensing. In "Cellular Standards" section of this article, we consider the progress in cellular standards in 5G and beyond; in "History of ISAC in 5G" section, we introduce the history of ISAC standardization in 3GPP and defined use cases in cellular; in "Current Standards Work on ISAC in 5G" section, we summarize the current standard work on ISAC in cellular; in "Channel Modeling Considerations" section, we discuss the ongoing channel modeling work for ISAC in 3GPP; in "Limitations of Current Work" section, we discuss the limitations on current work for ISAC; and "Future Directions" section summarizes potential future directions. Conclusions are drawn in the "Conclusion" section.

[1]Some parts of this image were generated using the Hugging face tool (https://huggingface.co).

ISAC IN WI-FI

In this section, we first summarize the research and standards activity, on sensing, and then discuss channel models and hardware impairments.

Progress in Wi-Fi Sensing

Due to their limited coverage range, Wi-Fi technology is generally suitable for personal area sensing and local area sensing use cases. Wi-Fi sensing usually involves a TX transmitting a Wi-Fi signal, and a RX using the received signal parameters from the signal preamble, such as received signal strength indicator (RSSI) and channel state information (CSI), to sense changes in the environment. Although these signal parameters are not exposed outside the Wi-Fi chip by most chip vendors currently, resourceful researchers have designed middleware packages [17] for some chips, to expose these parameters to applications. This led to a flourish in research, with numerous works exploring a myriad of sensing use cases, including respiration rate monitoring, people counting, localization, gesture recognition, fall detection, velocity estimation, object detection, etc. (see [5], [6], [18], and the references therein). Several open-source CSI datasets have also been made available to spur research [14]. Such solutions had to deal with the limitations of Wi-Fi hardware, which were designed for communication and not sensing. In particular, the measured CSI suffers from amplitude and phase errors, as elaborated later in the "Modeling of Hardware Impairments" section. Furthermore, although the software packages allowed accessing the CSI, there was no organic way to share the CSI across devices, coordinate sensing operation across multiple devices, etc. Most of these works also focused on sub-7 GHz Wi-Fi, where the limited available bandwidth and antenna array size limit the use cases.

To resolve these issues, work on the IEEE 802.11bf standard [13] for Wi-Fi sensing began in 2019, having concluded only recently in March 2025, and products are expected to roll out

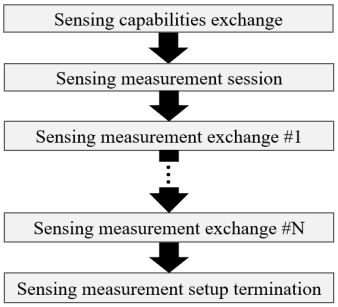


FIG. 4. Illustration of the steps involved in IEEE 802.11bf sensing protocol.

by 2026. In this standard, sensing uses the same hardware and similar signaling mechanisms defined for communication, but the sensing and communication exchanges are performed orthogonally in time. The overall sensing procedure defined by the standard is depicted in Fig. 4, and the steps are summarized as follows.

1) *Sensing Capabilities Exchange*: The participating Wi-Fi devices exchange their sensing capability information in this step.
2) *Sensing Measurement Session*: The participating Wi-Fi devices negotiate the parameters for a sensing session including selection of sensing duration and periodicity, type of sensing measurements, target transmit power, bandwidth, number of antennas to be used, etc., in this step.
3) *Sensing Measurement Exchange*: The sensing session involves multiple periodic measurement exchanges. The exchanges may be initiated by either an AP or a client station. Each such exchange may include a polling phase to poll the available RXs, a sounding phase where the sensing packets are transmitted by one or more TXs and are received by one or more RXs to obtain the sensing information, and a reporting phase where the RXs share information obtained with other devices.
4) *Sensing Measurement Setup Termination*: This step terminates the sensing session and is performed either explicitly or implicitly based on the negotiated session expiry timer.

The standard considers both: 1) sensing at sub-7 GHz bands, where bistatic sensing is supported; and 2) sensing at >45 GHz bands, where monostatic, bistatic, multistatic sensing, and coordinated sensing are supported. The standard also proposed passive sensing for >45 GHz bands, where the sensing device monitors frames transmitted by a TX that are not meant for sensing. To allow a client station to time multiplex bistatic sensing operation with multiple APs, the standard also defines mechanisms for the station to perform bistatic sensing with an AP with which it is unassociated. The standard also defined mechanisms for a client station to request an AP to perform sensing measurements on its behalf with other client stations, as a proxy. The standard also recommends mechanisms for Wi-Fi devices to reduce the amplitude and phase errors during the sensing session through implementation-dependent mechanisms. For example, amplitude errors may be reduced by not allowing an update of the RX automatic gain control (AGC) gain circuit during the sensing session. The standard considers both: 1) sensing at sub-7 GHz bands for use cases requiring coarse delay-angle resolution and wider coverage; and 2) sensing at >45 GHz bands for applications requiring high resolution and high accuracy. The key use cases of sub-7 GHz sensing and >45 GHz sensing are depicted in Table I [11]. For a more detailed overview of the IEEE 802.11bf standard, we refer the interested reader to [12].

TABLE I. Key use cases in IEEE 802.11bf standard

	Sub-7 GHz Sensing	>45 GHz Sensing
Key Use Cases	Presence detection, Proximity detection, Macroscale activity detection, Respiration rate detection, Fall detection, etc.	Gesture recognition, Pose estimation, Face recognition, 3-D-vision, Sneeze sensing, etc.

One key limitation of sub-7 GHz Wi-Fi sensing is the limited accuracy and generalizability of the solutions to different deployment scenarios. For example, it is found that the performance of respiration rate monitoring, people counting, fall detection, etc., can be significantly impacted by the presence of a moving fan or multiple moving targets. This is primarily due to the limited bandwidth and angular resolution at sub-7 GHz bands, which make it hard to filter out unwanted paths. However, with the recent release of Wi-Fi seven products in 2024 [19], there is scope for future work on fusing sensing information from 2.4, 5, and 6 GHz Wi-Fi bands (multilink operation) to improve the sensing accuracy and its generalizability. Although >45 GHz Wi-Fi sensing shows promise in generalizability, there are very few products on the market today that use >45 GHz Wi-Fi chips. To remedy this, IEEE has also started work on a new >42 GHz Wi-Fi standard IEEE 802.11bq in 2024, which plans to use a nonstand-alone deployment for >42 GHz bands, relying on sub-7 GHz bands for critical signaling [20]. Although this shows promise for adoption of >42 GHz Wi-Fi hardware and, correspondingly, sensing applications needing high accuracy, the products are not expected to hit the market in the next four years. An overview of the relevant Wi-Fi standards is provided in Table II.

Channel Models for Wi-Fi Sensing

Channel models are needed to generate CSI data, train different Wi-Fi sensing solutions, and evaluate their performance. However, modeling the CSI variation for various sensing applications is highly nontrivial. For example, it is very hard to predict the expected changes in the CSI when a person falls (fall detection use case). Consequently, research works have considered two approaches for getting channel data:

1) *Measured Data*: Some works have collected extensive Wi-Fi CSI data for different sensing use cases, with different subjects, measurement scenarios, etc. [14].
2) *Ray-Tracing Simulations*: Some works have generated a large amount of synthetic CSI data for different use cases

TABLE II. Wi-Fi Standards Relevant to Sensing

Standard	Definition	Freq. Bands	Key Features	Release Year
IEEE 802.11mc	Maintenance standard for main specification.	2.4, 5 GHz	Wi-Fi Round Trip Time (or FTM).	2016
IEEE 802.11ax	Main standard for Wi-Fi 6 and 6E products.	2.4, 5, 6 GHz	Support 6 GHz band, 160 MHz bandwidth, multiuser OFDMA, enhanced beamforming, target wake time, etc.	2021
IEEE 802.11be	Main standard for Wi-Fi 7 products.	2.4, 5, 6 GHz	320 MHz support, Multi-link Operation, restricted target wake time, multiple resource units, etc.	2024
IEEE 802.11bf	Standard to enable Wi-Fi sensing.	<7.125 GHz, >45 GHz	Trigger-based and nontrigger-based sensing, sensing-by-proxy, sub-7 GHz and >45 GHz solutions, etc.	2025
IEEE 802.11bn	Main standard for Wi-Fi 8 products.	2.4, 5, 6 GHz	Multi-AP coordination, seamless roaming, non-primary channel access, reconfigurable transceivers, peer-to-peer support, etc.	2028 (predicted)
IEEE 802.11bq	Standard for >42 GHz Wi-Fi.	>42 GHz	Nonstandalone mm-wave Wi-Fi operation.	Undecided

using ray-tracing simulations with approximate models for the sensing object and the environment [15].

The resulting CSI data have been used to test algorithms or train machine learning solutions.

Even on the standards side, the IEEE 802.11bf task group has proposed some channel models for sensing [21]. The purpose was to evaluate the efficacy of different proposals, e.g., the sensing waveform, rather than to evaluate solutions for any specific use case. Two scenarios were considered for the channel models: a living room of size $7 \times 7 \times 3$ m and a conference room of size $4.5 \times 3 \times 3$ m, with a device-free target moving at a speed of ≤ 1 m/s, as depicted in Fig. 5. The material properties of the different objects in the rooms can be found in [21] and [22]. The group adopted two channel models: 1) a ray-tracing channel model applicable to the >45 GHz band; and 2) a data-driven hybrid channel model for the sub-7 GHz and >45 GHz bands. The model is applicable both for monostatic and bistatic sensing. The ray-tracing model uses either a wooden block moving at the chosen constant velocity as a simple target to be sensed, or multiple scattering centers as a complex target to be sensed.

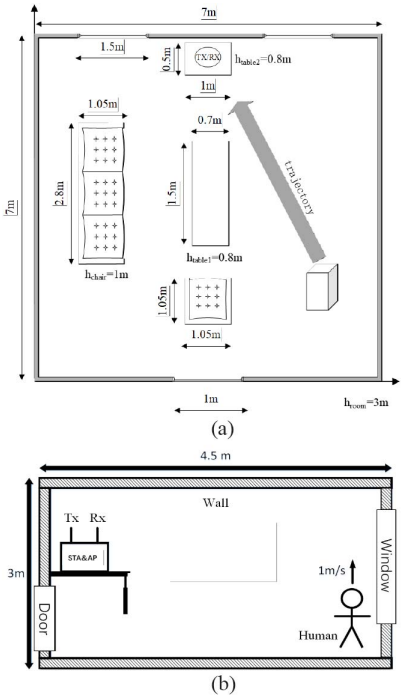


FIG. 5. Illustration of two scenario rooms considered for channel generation in IEEE 802.11bf [21]. (a) Living room. (b) Conference room.

As an example of the complex target, 17 scattering centers can be used to replicate the human body movement based on the Boulic model, which models the trajectory of 17 body joints as a person moves. Ray-tracing is then performed for the selected scenario to generate the channel impulse response (CIR) $h(t,\tau)$ for the target location at time t. Finally, a Fourier transform is taken across the delay domain τ to obtain the frequency response or CSI $H(t,k)$, where k is the subcarrier index. On the other hand, the data-driven hybrid channel model uses ray-tracing for generating the dynamic paths coming from the target, while using a stochastic channel model for the other nontarget paths. In particular, for the nontarget paths, first an initial CIR $\overline{h}_{\text{bgr}}(t,\tau)$ is generated for each time t using either the IEEE 802.11ax channel model (for sub-7 GHz bands) or the IEEE 802.11ay channel model (for >45 GHz band), where τ represents the channel delay. Then, an auto-regressive process is used to bring temporal correlation into the CIR as

$$h_{\text{bgr}}(t,\tau) = \begin{cases} \overline{h}_{\text{bgr}}(t-T_0,\tau), & \text{for } t \leq T_0 \\ \rho\overline{h}_{\text{bgr}}(t-T_0,\tau) + (1-\rho)\overline{h}_{\text{bgr}}(t,\tau), & \text{for } t > T_0 \end{cases} \quad (1)$$

where ρ, T_0 are hyperparameters determining the coherence time. In addition, these hyperparameters are adapted based on channel measurement data to better approximate the real channel in a specific scenario. Finally, the CIR from ray-tracing for target-related paths $h_{\text{tgt}}(t,\tau)$ is combined with $\tilde{h}_{\text{bgr}}(t,\tau)$ to generate the final CIR as

$$h(t,\tau) = h_{\text{tgt}}(t,\tau) + h_{\text{bgr}}(t,\tau). \quad (2)$$

A Fourier transform is then taken across the delay domain τ to obtain the CSI $H(t,k)$, where k is the subcarrier index. An

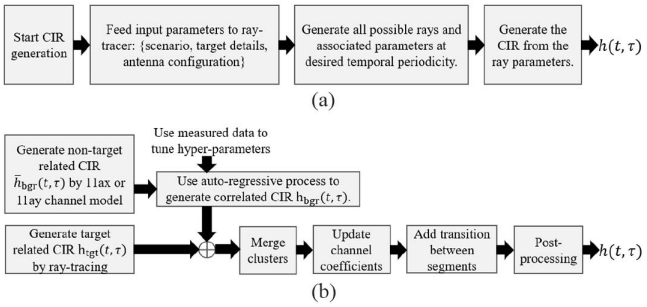


FIG. 6. Illustration of the channel generation processes considered IEEE 802.11bf task group. (a) Ray-tracing channel model. (b) Hybrid channel model.

overview of the two-channel models is depicted in Fig. 6. In comparison with ray-tracing, the hybrid channel model was found to better capture the sudden spikes in the CSI observed in measured data, potentially arising from unexpected scattering. The datasets from these channels can be found in [21].

Modeling of Hardware Impairments

Note that "Channel Models for Wi-Fi Sensing" section mainly describes the impact of the variation of the true physical channel on the CSI. However, apart from variations in the physical channel, Wi-Fi CSI is also subject to various transceiver hardware impairments that can impact sensing, especially for the bistatic case. Although the impact of some of these impairments such as AGC and symbol start-time error on CSI has been observed, and some postprocessing solutions for CSI have been proposed [23], [24], there is limited work on trying to characterize the behavior of the different impairments. Only recently, researchers have characterized these hardware impairments for some commercially available sub-7 GHz Wi-Fi chips [16], [25]. The key hardware impairments that impact CSI for sensing are elaborated as follows.

1) The received CSI at a Wi-Fi RX may experience abrupt amplitude variations, owing to changes in the gain provided by the AGC circuit. This gain change is common to all the subcarriers. Data from [16] show that this gain (in decibel scale) changes in increments of a chip-specific constant Δ, and the changes are independent for each receive antenna. An example of this gain error for a sample RX is depicted in Fig. 7(a), where we plot the change in CSI power (in dB) between successive received packets. The variation in the scatter plot comes from the difference in the AGC gain for the two packets, and it suggests a value of $\Delta = 0.5$.
2) A Wi-Fi RX detects the reception time of packets inconsistently, and this error in packet reception time δ leads to a phase ramp across subcarriers in the measured CSI. Data from [16] show that the time error is independently distributed for each packet, and its jitter across packets is smaller than 0.1 μs. An example of the distribution of this time error for a sample RX is depicted in Fig. 7(b).
3) The received CSI at a Wi-Fi RX may suffer from a discrete phase rotation, which is common across all subcarriers. This phase rotation can be due to cycle slips introduced by phase-locked-loop circuits or other hardware effects. Data from [25] show that these phase changes are independent for each packet and for each RX antenna and are multiples of a chip-specific constant Ψ. An example of this phase rotation ψ for a sample RX is depicted in Fig. 7(c), where we plot the inter-antenna phase difference of the CSI on a subcarrier as a function of time. The variation is due to different values of the phase rotations for the two RX antennas, and the variation suggests a value of $\Psi = \pi$ radians.
4) The crystals in Wi-Fi devices are nonideal and may have a mismatch of up to ± 20 parts-per-million (ppm). This error leads to: a) carrier frequency offset (CFO) between TX and RX; and b) oscillator phase noise at the TX and RX.[2] These impairments together cause a phase rotation ϕ of the CSI that is common to all receive antennas and subcarriers. Although the CFO changes slowly and may be assumed to be constant over a short sensing instance, the error from the phase noise accumulates over time, leading to an almost uniform distribution of this phase error ϕ between measurements separated by more than 10 ms in time. Thus, it is reasonable to model ϕ as a uniformly distributed random phase for each packet. An example of this phase rotation ϕ for a sample RX is depicted in Fig. 7(d), where we present a scatter plot of the difference in the CSI phase for two different packets at an RX antenna, as a function of the difference in reception time of the two packets. Here, the slope of the scatter plot represents the CFO value, and the width depicts the impact of residual phase noise as a function of time separation between the packets. As evident, ϕ for packets separated by more than 3 ms is almost independent and uniformly distributed.

It should be noted that several of these impairments do not exist in the case of monostatic sensing, such as the packet reception time error δ and the phase error ϕ from the CFO and the oscillator phase noise.

Based on these impairments discussed, the observed CSI between the mth transmit and nth receive antenna, on subcarrier k, and for the pth sensing packet can be modeled from the true channel CSI $H_{m,n}[p,k]$ (obtained from ray-tracing or other models in "Channel Models for Wi-Fi Sensing" section) as

$$\hat{H}_{m,n}[p,k] = 10^{\frac{x_{p,n}\Delta}{20}} e^{j(y_{p,n}\Psi + \phi[p])} e^{j2\pi k\delta[p]/T_s} H_{m,n}[p,r] \qquad (3)$$

where $x_{p,n}, y_{p,n}$ are random integers, $\delta[p]$ is the packet detection time error for packet p, $t[p]$ is the transmit time of packet p, $\phi[p]$ is phase rotation from CFO and phase noise for packet p, T_s is the OFDM symbol duration, and K is the number of

[2]There are also other effects of the crystal mismatch such as sampling rate offset and symbol timing offset. However, these have a relatively minor contribution to CSI [25] and hence are ignored here.

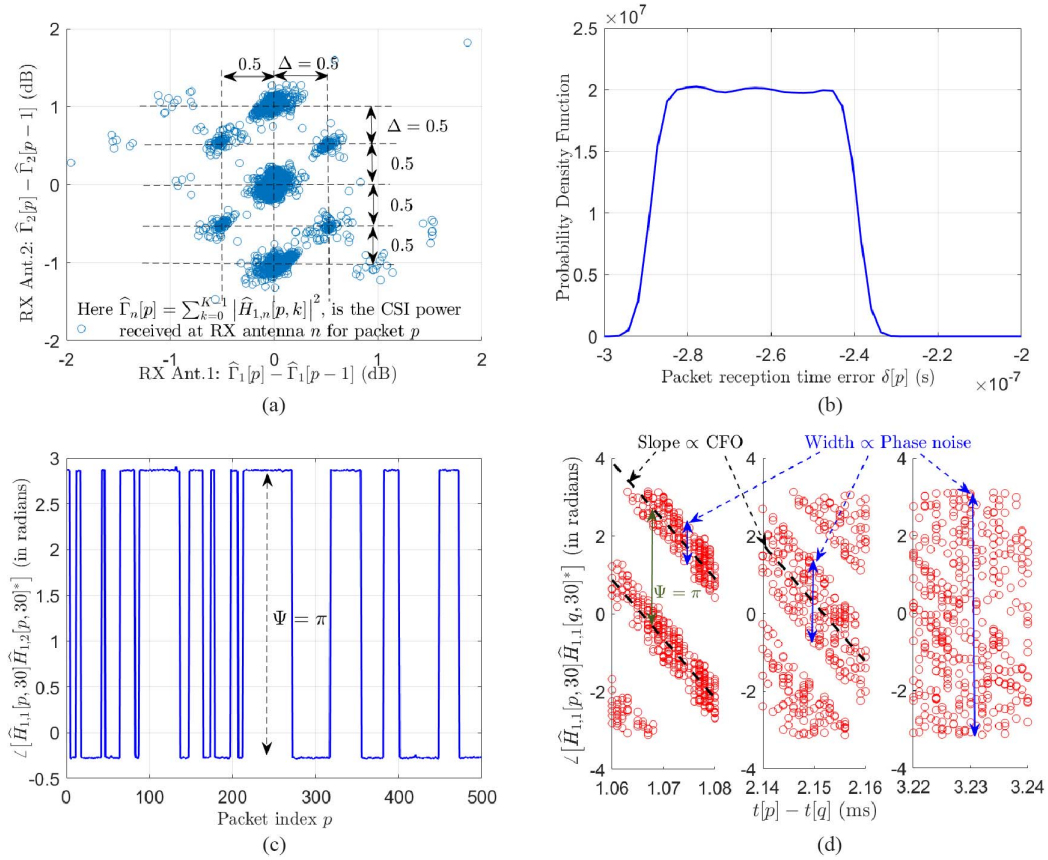


FIG. 7. Impact of the RX hardware impairments on the CSI measured at a 2-antenna RX for packets transmitted from a single antenna TX in a stationary channel. Both TX and RX are Intel AX210 Wi-Fi chips operating on the 5.7 GHz band. Here $\widehat{H}_{1,n}[p,k]$ is the measured CSI at RX antenna n on Sub-carrier k for packet p. (a) Scatter plot of difference in channel power $\widehat{\Gamma}_n[p] = \Sigma_k |\widehat{H}_{1,n}[p,k]|^2$ (in dB) between adjacent received packets $p-1, p$ at antenna 1 (on X axis) and antenna 2 (on Y axis). (b) Probability density function of the error in packet reception time δ across many received packets. (c) Inter-antenna CSI phase difference for subcarrier 30, over a sequence of received packets. (d) Scatter plot of the difference in the CSI phase at subcarrier 30 for two different packets at an RX antenna, as a function of the difference of reception time of the two packets. (a) AGC gain error. (b) Packet reception time error. (c) Discrete phase error. (d) CFO and phase noise error.

OFDM subcarriers. It is worth mentioning that (3) is a simplification under the assumption that CFO is a constant throughout the sensing session. Although this assumption is reasonable in many sensing applications, the CFO variation can play a significant role in applications where the power of sensing signal is orders of magnitude weaker than the nonsensing part of the CSI, such as in respiration rate monitoring. In such cases, a more involved expression for $\hat{H}_{m,n}[p,k]$ should be used. It should also be noted that some of the impairments discussed above may be reduced in IEEE 802.11bf compatible hardware. Thus, further analysis is required to characterize these impairments when 802.11bf compatible devices become available.

On the standards side, different IEEE task groups have proposed ways to model several transceiver hardware impairments for simulations. However, the focus was on impairments that can affect the task of communication, described as follows.

1) For the sub-7 GHz band, the IEEE 802.11ax task group proposed the use of an AM-AM RAPP model for the power amplifier nonlinearity, a fixed CFO value for the RX with respect to the TX, a phase noise model with a single zero and single pole transfer function for both the TX and RX [26].
2) For the >45 GHz band, the IEEE 802.11bf task group proposed the use of an AM-AM and an AM-PM RAPP model for the power amplifier nonlinearity, a fixed CFO value for the RX with respect to the TX, a phase noise model with a single zero and single pole transfer function for both the TX and RX, and an I-Q imbalance model [22].

Unfortunately, incorporating the above impairments into (3) is intractable. Fortunately, these impairments are expected to have a smaller impact on sensing, compared to those included in (3).

CELLULAR STANDARDS

History of ISAC in 5G

ISAC in 3GPP 5G systems refers to a technology that enables sensing services using the same new radio (NR) infrastructure as 5G communication networks. It allows the derivation of sensing information from both radio frequency (RF)-based (e.g., RADAR) and non-RF-based sensors [e.g., light detection and ranging (LiDAR) and time of flight (ToF) cameras], leveraging shared resources for dual purposes: communication and environmental awareness. The evolution of 3GPP

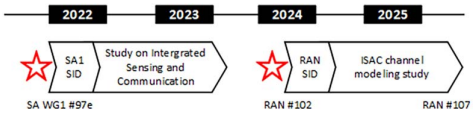


FIG. 8. Overview of standardization work of ISAC in 3GPP.

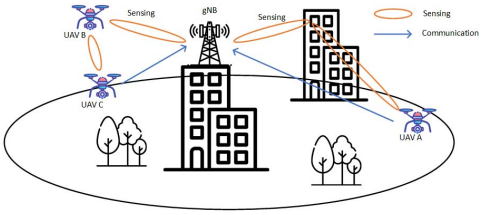


FIG. 9. Network assisted sensing to avoid UAV collision.

standards has marked significant milestones in ISAC as summarized in Fig. 8. These milestones reflect the industry's commitment to addressing technical challenges and enabling diverse applications for ISAC in both 5G and 6G systems. In February 2022, the initiation of the ISAC study item (SID) during the SA WG1 #97e meeting marked the beginning of ISAC standardization efforts. This study defines use cases and requirements for ISAC, covering both network-based and UE-based sensing capabilities. This study item aims to identify use cases and potential requirements for the enhancement of the 5G system to provide integrated communication and sensing services to address different target verticals and applications. By December 2023, the RAN #102 meeting launched the channel modeling study [27], which addressed gaps in existing channel models in [28]. The study focuses on defining channel modeling for object detection and tracking, targeting objects such as UAVs, humans, vehicles, and hazardous items across six sensing modes and frequencies from 0.5 to 52.6 GHz. It aims to identify deployment scenarios and refine existing models in [28], incorporating aspects such as radar cross-section (RCS), mobility, and spatial consistency. The discussions at RAN #105 meeting decided to explore potential extensions beyond channel modeling for ISAC, which has been an important channel model topic till now. In the March 2024 RAN #103 meeting, the progress of the ISAC study has been reported, with a focus on specific sensing targets (STs) and existing communication scenarios as a starting point, including UAVs, humans indoors/outdoors, automotive vehicles, automated guided vehicles, and objects creating hazards on roads/railways. In the recent 3GPP RAN #108 meeting, ISAC has been included in the scope of study on 6G radio, which means that ISAC will be regarded as one "Day 1" feature for 6G [29].

Key open issues include defining deployment scenarios and refining channel models in [28], incorporating aspects such as RCS, mobility, clutter/scattering patterns, and spatial consistency. In RAN #105 meeting, the group decided not to include additional studies other than channel modeling due to the time limit. During the whole period from RAN1 #116 to RAN1 #120b meetings, the two subtopics, ISAC deployment scenarios and ISAC channel modeling, have been thoroughly discussed. The 3GPP 6G workshop in Incheon, Korea (10–11 March 2025), highlighted 6G's role in enabling advanced services such as ISAC, extended reality (XR)/immersive communication, and artificial intelligence (AI)-driven applications. ISAC is envisioned to enhance the 6G network capabilities, providing a framework to integrate communication and sensing tasks. Key discussions focused on ISAC frameworks, multisource sensing data integration (RF/non-RF), and the need for technical studies in Release 20 to support diverse use cases. These milestones collectively highlight the progress made in ISAC standardization, positioning it as a cornerstone technology for future wireless networks, with applications spanning autonomous vehicles, smart cities, industrial automation, and beyond.

3GPP identified the key use cases and potential requirements for enhancement of the 5G system as the provisioning of sensing services addressing different target verticals/applications, e.g., autonomous/assisted driving, V2X, UAVs, 3-D map reconstruction, smart city, smart home, factories, healthcare, maritime sector [30]. The use cases examined in the report cover a wide range of applications, listed as follows.

1) Detection of objects and intruders in a smart home, on a highway, for railways, for a factory, and for predefined secure areas around critical infrastructure.
2) Collision avoidance and trajectory tracking of UAVs, vehicles, and autonomous ground vehicles (AGVs).
3) Automotive maneuvering and navigation.
4) Public safety search and rescue.
5) Rainfall and flood monitoring.
6) Health and sports monitoring.

As an example, Fig. 9 depicts the use case of "network-assisted sensing to avoid UAV collision." The commercialization of low-altitude UAVs is transforming industries such as logistics with applications in delivery, surveillance, and agriculture. With UAVs being deployed in so many industries, collision avoidance and effective management of UAV traffic are key challenges. 3GPP aims to leverage 5G connectivity and its wide coverage to realize real-time traffic management and collision avoidance for UAVs, addressing the key challenges in UAV deployment.

3GPP also identifies key performance indicators (KPIs) related to NR-based sensing (e.g., range, motion, and velocity) and performance requirements for the transfer of sensing data [30]. Performance requirements may differ for the different use cases, comprising both sensing and system KPIs listed as follows.

1) Sensing KPIs
   a) Accuracy of positioning estimate by sensing (for a target confidence level);
   b) Accuracy of velocity estimate by sensing (for a target confidence level);
   c) Sensing resolution, including range resolution and velocity resolution;

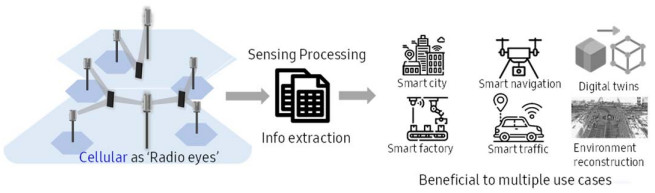


FIG. 10. Illustration of system architecture for ISAC.

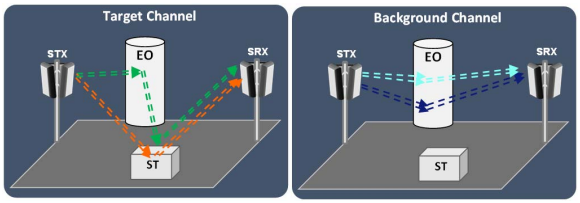


FIG. 11. Illustration of target channel and background channel.

d) Maximum sensing service latency.

2) System KPIs
   a) Refreshing rate;
   b) Missed detection rate;
   c) False alarm rate.

These KPIs ensure that ISAC meets the diverse requirements of various applications identified by 3GPP, balancing communication and sensing efficiency.

Current Standards Work on ISAC in 5G

Although notable progress has been made in 3GPP standardization, ISAC standardization remains in its initial phase. Current efforts are strategically divided between two key working groups:

1) *SA1 Working Group*: This group drives service requirement formulation through use case categorization and KPI derivation.
2) *RAN1 Working Group*: This group focuses on advancing ISAC-specific channel modeling with multiphysics considerations.

In terms of system architecture, multiple sensing modes have been defined in 3GPP, which can be categorized into two types: monostatic sensing and multistatic sensing (of which bistatic is a special case), as shown in Fig. 2. Monostatic sensing requires a single Next-generation Node B (gNB) for operation, while multistatic sensing requires at least two gNBs. The ISAC system architecture, which incorporates these sensing modes, is presented in Fig. 10. The sensing data collected by the ISAC system undergo processing to form sensing datasets. These datasets enable various innovative applications, such as smart traffic management and environmental reconstruction.

One major ongoing work in 3GPP related to ISAC is the channel modeling for ISAC. The channel modeling is based on the statistical model specified in [28]. However, taking the sensing channel into account, new features are applied to the models, including:

1) Separate the generation of the target channel and the background channel.
2) Target modeling, including the target object's RF energy scattering characteristics, e.g., RCS, and the number of scattering points (SPs).
3) Environment object (EO) modeling, including type-1 and type-2 EOs.

The detailed description for ISAC channel modeling can be found in the "Modeling of Hardware Impairments" section.

Channel Modeling Considerations

The ISAC channel model integrates sensing and communication by distinguishing between STs and environmental objects. Unlike conventional communication-centric models, ISAC requires a hybrid channel architecture that simultaneously optimizes data transmission and environmental sensing. This model combines a target channel (containing multipath components influenced by specific STs) and a background channel (other multipath signals) as shown in Fig. 11. This model integrates two distinct components: 1) the target channel, which captures multipath effects influenced by STs; and 2) the background channel, which accounts for environmental multipath signals unrelated to STs.

The channel parameters for the links from the sensing transmitter (STX) to ST and ST to the sensing receiver (SRX) are inherited from existing communication channel models. This model assumes that each ST acts as either a TX or a RX in its respective link. Multiple STs can be sensed simultaneously, but paths interacting with multiple STs are excluded if blockage/forward scattering is disregarded [31]. The target channels for $ST_q, (1 \le q \le Q)$ include all impacted multipath components, which may interact with the stochastic clusters in STX-ST or ST-SRX links. The ISAC channel coefficient $H(t,k)$ at time t and on subcarrier k can be represented as

$$H(t,k) = H_{\text{tgt}}(t,k) + H_{\text{bgr}}(t,k) \tag{4}$$

where H_{tgt} is the target channel which combines line-of-sight (LOS) and non-line-of-sight (NLOS) components modified by ST interactions, while H_{bgr} is the background channel which captures the channel components contributed by the environmental scatters. The framework maintains compatibility with standard 3GPP models while enabling simultaneous sensing-object detection and communication through this dual-channel structure.

For the target channel, a two-segment channel generation method is used as shown in Fig. 12 to model interactions between STXs, STs, and SRXs. The first segment, STX-to-ST, models propagation from the TX to the target, while the second segment, ST-to-SRX, captures propagation from the target to the RX. These segments are concatenated by summing delays and multiplying power (in linear domain), where the total power incorporates the ST's RCS effects. Two options are available to concatenate the segments:

1) *Option 1*: Each NLOS ray in the TX-ST link is coupled with each NLOS ray in the ST-RX link, resulting in a full combination of all possible paths.

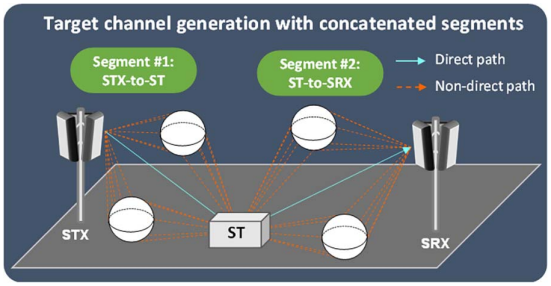


FIG. 12. Illustration of target channel generation by concatenating two segments, i.e., segment #1 STX-to-ST and segment #2 ST-to-SRX.

2) *Option 2*: The NLOS rays in the TX-ST link are 1-by-1 randomly coupled with the NLOS rays in the ST-RX link. If the number of rays in the two links (e.g., M1 and M2) is unequal, min(M1, M2) rays are randomly selected from the link with the number of rays for the coupling operation.

For the background channel, the generation methods are different for different sensing modes. For bistatic modes [transmission and reception point (TRP)-TRP, TRP-UE, UE-TRP, and UE-UE], the background channel is generated by using the cluster-based method in [28] with the incorporation of the absolute time-of-arrival. For monostatic modes (TRP monostatic sensing and UE monostatic sensing), reference points (RPs) are introduced to serve as virtual RXs specifically for background channel modeling. This ensures the generated channel matches the background channel derived from measurements or ray-tracing results. The background channel is generated between STX/SRX and the RPs using the cluster-based method in [28] with modifications on making the arrival angles and departure angles equal and applying the absolute time-of-arrival.

As a critical component, in addition to traditional channel models, the RCS is introduced to quantify the hypothetical area of a target that isotropically scatters incident radar power to match the observed RX power density. The value of RCS is dependent on material properties and angle-dependent scattering characteristics, reflecting path-loss variations and multipath fluctuations across different geometries. For example, metallic UAVs exhibit higher RCS values than human targets due to structural reflectivity. For RCS modeling, we have the following:

$$\sigma_{\text{RCS}} = \sigma_M \sigma_D \sigma_S \tag{5}$$

where σ_M denotes a large-scale parameter that captures the mean RCS of scattering points of the ST (SPST), σ_D denotes a small-scale parameter that is fixed to 1 if the ST is of small size; otherwise, it is angle-dependent. σ_S denotes a small-scale parameter that follows a log-normal distribution with mean μ_{σ_S} and standard deviation σ_{σ_S}.

The ISAC physical object model categorizes entities as STs or EOs. As objects of interest, STs such as UAVs, vehicles, or humans are modeled with consideration of the properties including the number of SPs, location, RCS, and polarization. In addition to STs, nontarget entities or EOs are also

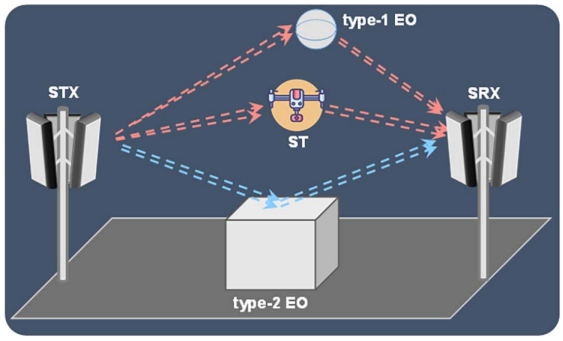


FIG. 13. Illustration of type-1 EO and type-2 EO.

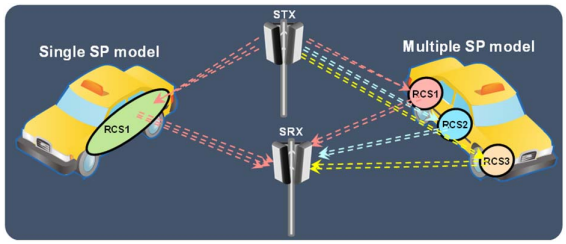


FIG. 14. Illustration of single SP model and multiple SP model.

introduced to better reflect the effect of the environmental scatters in the real environment. EOs are categorized into two types as shown in Fig. 13, as follows.

1) *Type-1 EO*: They behave similarly to STs but are not targets of interest, e.g., animals and trees.
2) *Type-2 EO*: They represent large environmental structures (e.g., walls and buildings) modeled as finite reflective surfaces that exhibit specular reflection.

In terms of the number of SPs being used to model the ST, two RCS models are defined, as shown in Fig. 14. These include the single SP model and the multiple SP model, which are defined to support diverse sensing scenarios, balancing deterministic and stochastic parameters while accommodating object-specific geometries (vehicles, UAVs, humans).

1) *Single SP Model*: Targets, for e.g., UAVs and humans, can be modeled by a single SP.
2) *Multiple SP Model*: Targets, for e.g., vehicles and AGVs, can be modeled by multiple SPs where the RCS values of each SP are individually determined. Vehicles with multiple SPSTs have five specific points: front, left, back, right, and roof.

Limitations of Current Work

As discussed in previous sections, the standardization of ISAC within the 3GPP framework remains in its early stages. Although a growing number of ISAC trials and testbeds have been demonstrated, current implementations largely adopt simple designs. Specifically, sensing and communication functionalities are often executed in a time-domain multiplexed manner, where the two operations operate sequentially rather than concurrently. This fragmented approach limits the potential for true synergy between sensing and communication,

underscoring the need for more advanced integrated architectures to fully unlock the capabilities of ISAC systems. Certain existing ISAC implementations allocate dedicated resource blocks with sensing-optimized reference signals (RSs) for sensing functionalities. However, such designs often require reserving $> 15\%$ of communication-dedicated resources for sensing tasks [32]—a configuration that may prove suboptimal in future 6G systems. This substantial allocation risks creating resource competition between sensing and communication operations, potentially compromising spectral efficiency and scalability in ultrahigh-capacity 6G networks.

Future Directions

In the ITU-R recommendations for the future 6G framework [33], ISAC is identified as a new usage scenario for IMT-2030. Sensing-related capabilities, such as object detection, localization, imaging, and mapping, are considered essential for IMT-2030. In 6G, new sensing scenarios will be defined to further leverage the capabilities of sensing and communication through ISAC. Specifically, for communication-assisted sensing, cellular systems will provide sensing-related services. Meanwhile, in sensing-assisted communication, sensing information, including environmental and radio channel conditions, will be utilized to improve communication performance. In this way, a synergistic effect can be expected for future 6G systems.

In the context of 6G systems, key research efforts in the physical layer will focus on designing a unified architecture that can simultaneously support communication and sensing functionalities. Potential research directions in this domain can be summarized as follows.

1) *Waveform Design*: Optimal waveforms for communication and sensing, respectively, exhibit significant differences, necessitating innovative approaches for a joint design. A key challenge lies in designing harmonized waveforms and achieving a unified design that can effectively support both functionalities. Addressing these design considerations will be critical to advance the capabilities of 6G ISAC systems.
2) *Design of Sensing RS*: Enabling dual functionality for the sensing RS will be a priority for 6G ISAC, ensuring high sensing accuracy to enable advanced use cases, e.g., digital twin, while minimizing communication overhead via joint waveform optimization, shared resource allocation, and interference mitigation.
3) *Cooperative Sensing*: Mechanisms for cooperative sensing can be developed in which cooperation between multiple sensing nodes can be used to improve the sensing performance.
4) *Multimodal Sensing*: It has the potential to further enhance the sensing performance, especially sensing accuracy, via integrating complementary sensing data from various sources (e.g., radar, video, and ultrasound) to mitigate single-source limitations, reduce sensing uncertainty, and improve robustness.
5) *Interference Management*: For multistatic sensing modes, there will be interference between the communication link and the sensing link. Developing robust interference mitigation mechanisms, therefore, emerges as a critical enabler to optimize the synergistic performance of sensing and communication in 6G ISAC systems.

From a network perspective, the research will focus on developing a suitable network architecture suitable for 5G-A and 6G ISAC. Potential research directions in this domain can be categorized as follows.

1) *Network Architecture for Sensing-Based Services*: Future networks will natively support diverse sensing modes and heterogeneous service requirements through the evolution of the protocol stack. This necessitates the introduction of dedicated sensing-plane functions within the core network architecture. Key architectural challenges center on two aspects: Optimal functional decomposition between the radio access network and the core networks, and interface design for cross-domain information exchange.
2) *Protocol Optimization for Sensing-Native Network*: Within a network-centric optimization framework, ISAC-aware protocol stacks require cross-layer enhancements spanning dynamic sensing parameter configuration, low-latency measurement reporting mechanisms, predictive mobility management, etc.
3) *Resource Allocation and Optimization*: Further optimizations for resource allocation will also be important for a better balance between communication and sensing.

CONCLUSION

The integration of sensing and communication capabilities through ISAC represents a pivotal advancement in wireless systems, enabling new transformative applications. This article has provided an overview of ISAC standardization efforts within key frameworks, including Wi-Fi and cellular communication.

On the Wi-Fi side, the new IEEE 802.11bf standard provides frameworks for monostatic, bistatic, and multistatic sensing, with the technology covering both the sub-7 GHz and > 45 GHz bands, and products expected to roll out in 2026. Significant work has also been done to adapt existing channel models to enable their use for ISAC, through the use of ray-tracing. The article also underscores the limited work done in characterizing the transceiver hardware impairments and their impact on ISAC and provides some initial progress on the same.

On the cellular side, the system architecture considers monostatic, bistatic, and multistatic sensing. Significant progress has been made in identifying key use cases and KPIs to achieve them. On the channel modeling side, separate generation of the target channel and background channel is proposed, which includes modeling of the target scattering and its radar cross section, modeling of EOs, etc. Looking ahead, future

directions must prioritize innovations at the physical layer, such as advanced waveform design and cooperative sensing, alongside network-level optimizations to enable scalable and efficient ISAC systems.

The continued evolution of ISAC standards will be instrumental in shaping the future telecommunication ecosystem, where ISAC will underpin critical applications, from autonomous networks to intelligent environments, enabling even more productive use of the wireless infrastructure and the electromagnetic spectrum.

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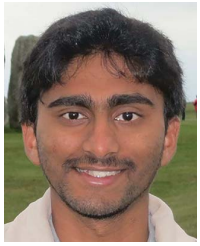
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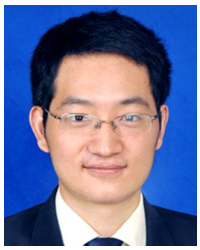
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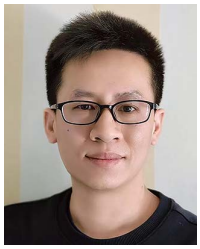
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